

Contemporary Technology Development (Industry Case Study Based)

Module 2: Case Study 1 – Renewable Energy Technology

2 Marks Questions with answers

1. Define renewable energy.

Renewable energy is energy generated from natural sources that are replenished constantly, such as sunlight, wind, water, and geothermal heat.

2. What is the main working principle of photovoltaic solar cells?

PV cells convert sunlight directly into electricity using the photovoltaic effect in semiconductor materials like silicon.

3. Name two types of wind turbines.

Horizontal Axis Wind Turbine (HAWT) and Vertical Axis Wind Turbine (VAWT)

4. What is the function of an inverter in a solar power system?

It converts the DC (direct current) produced by solar panels into AC (alternating current) used by household appliances.

5. Mention any two types of energy storage systems.

Lithium-ion battery (electrochemical) and Pumped hydro storage (mechanical)

6. Define a solar farm.

A large-scale installation of solar PV panels designed to generate electricity and supply it to the grid.

7. What is the purpose of the National Solar Mission in India?

To promote solar energy development and achieve large-scale deployment of solar power to reduce dependency on fossil fuels.

8. Mention one advantage and one limitation of solar farms.

Advantage: Clean energy generation.

Limitation: Land acquisition and maintenance issues.

9. What is intermittency in renewable energy?

Variability in power generation due to weather or time, especially in solar and wind power.

10. What is agrivoltaics?

Dual use of land for both agriculture and solar panel installation.

11. What role does energy storage play in renewable energy integration?

It helps manage intermittency by storing excess energy and releasing it when generation is low.

12. . What challenges do solar and wind energy face?

Key challenges include energy storage, intermittency, and grid integration. Solutions like advanced batteries and improved energy forecasting are helping overcome these hurdles.

5 Marks Questions with answers

1. Explain the components of a solar PV system.

A typical solar PV system includes:

- **Solar Panels:** Capture sunlight and generate DC power.
- **Inverter:** Converts DC to AC.
- **Mounting Structure:** Supports the panels.
- **Charge Controller:** Protects the battery in standalone systems.
- **Battery Storage (optional):** Stores energy for later use.

2. Explain the technical components of a solar farm.

- Solar PV modules
- Mounting structures
- Inverters and transformers
- SCADA systems
- Transmission lines and substations

3. What are the benefits of large-scale solar farm development in India?

- Reduction in carbon emissions
- Increased energy security
- Employment generation
- Support for renewable energy goals

4. Explain how renewable energy technologies are lowering electricity costs globally.

- Mass production reduces cost per unit
- Technological advancements increase efficiency
- Competitive tariffs (solar auctions)
- No fuel cost (unlike coal/gas plants)

5. What is the role of smart grids in integrating renewable energy into power systems?

- Real-time monitoring and control
- Demand-response mechanisms
- Load balancing during intermittent supply
- Enables integration of storage and distributed generation
- Growth in solar, wind, and battery industries

6. Analyse the economic, social, and environmental benefits of transitioning to renewable energy.

Economic: Jobs, investment, cost savings

Social: Energy access, health improvement, equity

Environmental: Emission reduction, resource conservation

- Long-term sustainability of energy supply
- Net-zero and Paris Agreement commitments
- Supportive regulations and subsidies

7. What are the examples of renewable resources & non-renewables resources?

The examples of renewable resources are:- Geothermal, solar, wind, hydroelectricity, and biomass are all examples of renewable energy sources. Biomass has the largest contribution, accounting for 50%, followed by hydroelectricity (26%), and wind power (18%). Geothermal energy is created by capturing the natural heat of the Earth.

Crude oil, natural gas, coal, and uranium are examples of non-renewable resources. These are all resources that are converted into commercially viable products. The fossil fuel industry, for example, extracts crude oil from the ground and transforms it into gasoline.